

Neural SC Decoder for Two-User MAC Polar Codes — Executive Summary

Problem

Two users transmit polar-coded messages through a Gaussian MAC: $Z = (1-2X) + (1-2Y) + \text{noise}$. We replace the analytical SC decoder’s tree operations with learned neural networks (weight-shared MLPs, ~25K params, 4-class joint (u,v) output). Goal: match analytical SC BLER at all code lengths N.

Architecture

The decoder walks the polar code factor graph, using learned MLPs for each operation: CalcLeft (f-node), CalcRight (g-node), CalcParent (gated residual), and embedding/decision heads. All ops are weight-shared across tree positions — a model trained at one N works at any N. Training uses sequential teacher-forced tree walk with curriculum learning (N=16 -> 32 -> 64 -> ...).

Results (GMAC, SNR=6dB, Class B symmetric rate)

N	SC BLER	NN-SC BLER	Ratio	Verdict
32	0.046	0.046	1.0x	Matches SC
64	0.025	0.026	1.03x	Matches SC
128	0.016	0.019	1.17x	Close
256	0.005	0.019	3.7x	Gap
512	0.001	0.045	40x	Fails

On discrete BEMAC channel: NN **beats** SC at all N (including 0.5x at N=64). Neural SCL(L=4) beats analytical SCL(L=4) by up to 8x.

What Works / What Fails

Works: Curriculum learning, gated residual CalcParent, stable cosine LR (improved N=128 from 1.69x to 1.17x), Neural SCL at N≤64, MC frozen set design for Class B.

Fails: Scaling to N≥256. Root cause: weight-shared MLPs accumulate per-node errors through 8+ tree levels. Tried: larger model (d=32, 157K params — matches SC at N=32 but fails at N≥64 due to training budget), per-level ops (solves curriculum transfer but slow), NPD-style fast_ce parallel training (4-class MAC joint structure doesn’t decompose through binary sign-flips; two-decoder binary approach impossible at Class B rates where $R_u > \text{marginal capacity}$).

Key Open Problem

The single-user Neural Polar Decoder (Aharoni et al.) scales to N=1024 using $O(\log N)$ gradient depth via parallel teacher-forced training. Adapting this to the MAC’s joint 2x2 probability structure is the critical unsolved problem. The MAC requires joint processing of both users — neither independent per-user decoders nor naive 4-class formulations work. A correct MAC-adapted parallel training method would likely solve the N-scaling bottleneck.

Code: <https://github.com/eitanspi/polar-codes-mac> | **Models:** d=16 checkpoints for N=32-1024 |
Data: 824+ simulation points